

Topic:-

Visit the water testing laboratory in your area and collect information about tests for checking water quality.

Introduction:-

Water is the elixir of life. Clean drinking water is essential for the survival and good health of every human being, animal, and plant. In our modern cities and villages, natural water sources like rivers, wells, and borewells are facing increasing contamination from untreated industrial effluents, sewage wastewater, and agricultural pesticides.

Drinking contaminated water causes deadly water-borne illnesses such as cholera, typhoid, jaundice, and amoebic dysentery. To ensure that municipal tap water and groundwater are safe for drinking, municipal corporations and public health departments establish specialized Water Testing Laboratories. For our Class 8 science project on Pollution, our student team visited a district water quality testing laboratory to observe the scientific tests conducted to check water purity.

❖ Three Major Categories of Water Impurities:-

- **Physical Impurities:** Mud, silt, floating debris, bad color, and foul odor.
- **Chemical Impurities:** Excess salts, nitrates, fluorides, lead, and acidic pH.
- **Biological Impurities:** Harmful bacteria like E. coli and coliform pathogens.

Aim & Lab Visit Details:-

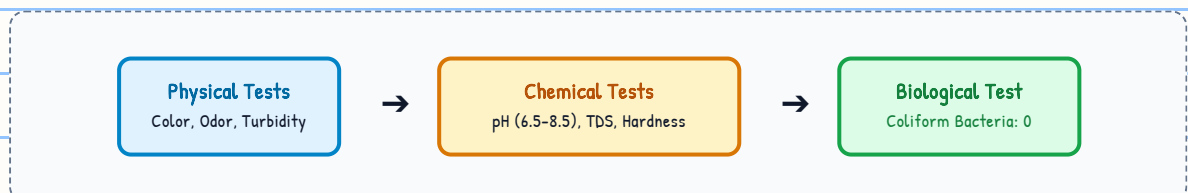
Aim: To visit the municipal water testing laboratory, interview water quality analysts, and document physical, chemical, and biological water testing procedures.

❖ Objectives of the Study:-

- To understand standard parameters of potable (safe drinking) water.
- To see how digital pH meters, TDS meters, and turbidity meters operate.
- To learn how the bacteriological coliform test detects sewer contamination in water pipes.

❖ Testing Instruments Inspected in the Lab:-

- Digital pH meter, Nephelometric Turbidity meter, and Electrical Conductivity (TDS) meter.
- Spectrophotometer, burettes for titration, and bacteriological incubator (37°C).



Standard Water Testing Procedures:-

The water laboratory chemist demonstrated four essential diagnostic tests:

- **1. Testing pH Level:** A calibrated glass electrode is immersed in the water sample. Pure drinking water must have a neutral pH between 6.5 and 8.5. Water below 6.5 is acidic and corrodes pipes; water above 8.5 is overly alkaline.
- **2. Total Dissolved Solids (TDS Test):** The digital TDS meter measures dissolved mineral ions in parts per million (ppm). Desirable drinking water has a TDS between 100 and 300 ppm. TDS above 500 ppm tastes salty and causes kidney stones.
- **3. Water Hardness (EDTA Titration):** Measures dissolved calcium and magnesium salts. Hard water does not form lather with soap and creates white scaling in kettles.
- **4. Coliform Bacteria Test (MPN Method):** Water is added to lactose broth tubes and incubated at 37°C for 24 hours. Production of gas indicates E. coli bacteria from sewage!



BIS Drinking Water Standards Table:-

The table below lists the Bureau of Indian Standards (IS 10500) parameters explained to us:

<i>Water Quality Parameter</i>	<i>Acceptable BIS Limit</i>	<i>Permissible Upper Limit</i>	<i>Effect If Exceeded</i>
pH Value	6.5 to 8.5	No relaxation	Corrosion of metal pipes / Bitter taste
Turbidity (NTU)	1.0 NTU	5.0 NTU	Cloudy water; shields germs from chlorine
Total Dissolved Solids (TDS)	Up to 300 mg/L (ppm)	500 mg/L (ppm)	Salty taste, stomach irritation, scale
Total Hardness (as CaCO ₃)	200 mg/L	600 mg/L	Soap does not lather; boiler scaling
E. Coli / Coliform Bacteria	Must be Zero (0) in 100 mL	Zero (0)	Severe diarrhea, cholera, and typhoid!

Scientific Analysis & Discussion:-

Analyzing our lab visit data revealed key scientific principles of environmental sanitation:

- **Why Chlorination Is Vital:** Municipal treatment plants add chlorine gas or bleaching powder (1 to 2 ppm) to water. Free chlorine oxidizes the cell walls of bacteria, killing them within 30 minutes and providing residual disinfection in pipelines.
- **Turbidity Blocks Disinfection:** When river water is muddy (high turbidity), suspended clay particles wrap around bacteria and shield them from chlorine rays. Water must be treated with alum (phitkari) for coagulation before filtration.
- **Difference Between RO Water and Boiled Water:** Reverse Osmosis (RO) filters remove excess dissolved minerals (TDS). Boiling kills 100% of biological germs but does not reduce TDS.

❖ Why Zero Coliforms Is Non-Negotiable:-

Even a single coliform bacterium detected in 100 mL of tap water indicates that raw sewage has leaked into underground drinking water lines!

Everyday Applications & Safe Drinking Tips:-

Our visit highlighted practical rules every household should follow to ensure safe water:

- **1. Annual Water Testing:** Families using borewells or well water should send a sample to the government water testing laboratory once every year before the monsoon.
- **2. Boiling During Monsoons:** Bring drinking water to a rolling boil for 10 minutes. This is the cheapest and most reliable way to sterilize germs.
- **3. Cleaning Overhead Water Tanks:** Household water storage tanks must be drained, scrubbed with bleaching powder, and washed at least once every three months.
- **4. Clean Handling at Home:** Never dip bare fingers into water pots; always use a clean handled ladle (kalashi) to draw water.

❖ Testing Kit for Villages:-

The chemist showed us portable field test kits (Jal-TARA) used by village panchayats to check water quality in remote schools for under ₹5 per test.

What We Learned & Conclusion:-

- Potable drinking water must be physically clear, chemically balanced (pH 6.5-8.5, TDS < 300), and 100% germ-free.
- Water testing laboratories use sensitive instruments like digital pH meters, TDS meters, and coliform incubators.
- Chlorination and boiling provide reliable shields against water-borne epidemic diseases.
- Regular testing of community water sources prevents pollution-related health crises.

Conclusion:-

Our visit to the district water testing laboratory provided a thorough practical understanding of water pollution and quality control. We saw how scientific testing protects thousands of families from hidden water contaminants. Clean drinking water is a fundamental human right, and scientific testing ensures that every drop from our taps is safe and life-giving.

References:-

- Maharashtra State Board Class 8 General Science Textbook (Chapter 8: Pollution).
- Bureau of Indian Standards (BIS) – Drinking Water Specification (IS 10500:2012).
- Laboratory Manual and Field Log from District Water Quality Testing Laboratory.